Open problems:
The Born rule is equivalent to the von Neumann entropy

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

No item below blocks the verification of a main claim of the brief.

From the brief

1. **Infinite entropy.** EQST takes the entropy to be real-valued, but in infinite dimension the von Neumann entropy of some density operators, and the Shannon entropy of some countable families of weights, is $+\infty$. With real values BR-ENT and [B]MUTEX cannot hold in infinite dimension. Which codomain is intended, and whether [B]MUTEX should be restricted to ensembles of finite entropy, is open. The formalization uses $[0, \infty]$. (Brief, Sec. 3, EQST; Log, "Problems found", first item.)

2. **Independence of the conditions.** The conclusion says that the only extra requirement is the consistency [B]MUTEX. Whether BR-BORN and BR-ENT are each independent of the other over EQST (without [B]MUTEX), that is, whether there are a p satisfying EQST and BR-BORN and an S violating [B]MUTEX, and conversely, is not addressed. (Brief, Sec. 4.)

From the formalization

1. **Countable affinity and Corollary 8.** EQST is formalized with affinity for mixtures of two pure states only. The brief's affinity for countable mixtures, and Corollary 8 (that p is determined by its values on pure states, which in infinite dimension needs countable mixtures and so countable affinity), are not formalized. (Log, "Differences", third item.)

2. **The Bloch-ball picture.** The geometric diagonalization of Fig. 1 and eqs. (4)–(5), and the picture of $p(\psi|\cdot)$ as an affine function on the Bloch ball of Fig. 3, are not formalized; the proofs are algebraic instead. (Log, "Differences", last item.)